

JAGADEESWARI J M

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EXECUTIVE SUMMARY

Data Science graduate and Data Associate skilled in Python, SQL, Machine Learning, VBA, and Data Visualization. Experience building predictive models, automating Excel-based business systems (VBA, Power Query), and developing real-time dashboards. Strong problem-solver with hands-on project and industry experience, eager to apply data-driven insights to support business decision-making and operations.

EDUCATION

Master of Science in Data Science — Annamalai University, Chidambaram, Tamil Nadu

Expected Graduation: May 2027 (Present)

Bachelor of Science in Data Science — The American College, Madurai, Tamil Nadu

Graduated: May 2025 | GPA: 8.7

Relevant Coursework: Machine Learning, Data Visualization, Deep Learning, Python & R Programming, SQL, Big Data Analytics

SKILLS

Programming: Python, R, SQL, Java, VBA

ML/AI: Machine Learning, Deep Learning, Predictive Modeling, Classification, Regression, Feature Engineering, Neural Networks (TensorFlow, PyTorch)

Data Processing & Automation: Pandas, NumPy, ETL, Data Cleaning, Data Wrangling, Excel VBA Automation, Power Query

Databases: MySQL, PostgreSQL

Visualization & Reporting: Power BI, Tableau, Matplotlib, Seaborn, Plotly, Advanced Excel

Deployment & Tools: Streamlit, Model Deployment, Jupyter Notebook, Git/GitHub

EXPERIENCE

Data Associate Apprentice — Aero Threads — Pudukkottai, Tamil Nadu | *Jun 2026*

- Built and maintained a 55-vehicle fleet maintenance tracking system in Excel (VBA + Power Query), including automated service log validation, edit/delete workflows, and alert systems.
- Designed and corrected formula logic across Live_Alerts, Supervisor_Report, and DB_Vehicles sheets, ensuring accurate sync between service history and vehicle records.
- Automated invoicing workflows in a multi-user Excel system (Bill_checking) using formula-based logic, improving accuracy without disrupting existing team workflows.
- Developed an admin monitoring workbook and a Fleet Trips Dashboard using SUMPRODUCT-based aggregation for operational reporting.

Data Science Research Associate — PRCBO Tech Hub — Remote | *May 2026 – Jun 2026*

- Developing custom analytical dashboards and automated data pipelines using standard ES6 JavaScript and HTML5, managing Git repository standardizations.
- Designing and executing client-facing visualization dashboards using HTML5/CSS3 variables and vanilla JavaScript ES6, reducing UI render latency by 25%.
- Instituted formal repository workflows unified on Conventional Commit messaging, reducing integration conflict rate.
- Conducted statistical evaluations and analytical models on public datasets for university-level presentations and research modules.

Data Science Intern — myplaybook Professional Services — Remote | *Nov – Dec 2025*

- Engineered Python automation scripts and performed extensive exploratory data analysis and statistical profiling.

Machine Learning Intern — Sebot Systems — Remote | *Nov – Dec 2024*

- Developed an end-to-end customer churn prediction pipeline utilizing XGBoost and Random Forest models.

Data Science & Machine Learning Intern — Edureka Enterprise | Dec 2023

- Worked on predictive modeling, data preprocessing, and ML pipelines.
- Built end-to-end data workflows and contributed to real-world analytics tasks.

Big Data Engineering Intern — IIT Allahabad (Hadoop Core) | May – Jun 2024

- Processed 500GB+ datasets using Hadoop MapReduce; optimized ETL pipeline reducing query execution time by 35%.
- Implemented batch processing, reducing computational overhead by 40%.

Data Analytics Intern — Intellizy Studio | Jun – Jul 2024

- Applied data analytics techniques to business use cases.
- Performed EDA, reporting, and dashboard creation for real datasets.

IoT & Robotics Intern — Corizo | Jul – Aug 2024

- Worked on IoT system integration, robotics components, and sensor-driven data analytics.

PROJECTS & RESEARCH PAPERS

Physics-Informed Machine Learning for Tire Degradation and Lap-Time Risk Modeling in Motorsport

- Won Best Paper Award at a national conference; developed an interpretable ML framework integrating domain physics with regression and classification models, achieving $R^2 = 0.96$ and Macro-F1 = 0.99.

Physics-Informed Neural Networks for Lap-Time Optimization and Fuel Strategy in Formula Racing

- Research paper submitted to ICRTMS-2026, Thiruvalluvar University; extends prior tire degradation research to lap-time and fuel strategy optimization.

Hybrid Intelligence for Autonomous Vehicles

- Developed end-to-end autonomous vehicle decision system using YOLOv8 object detection, LSTM trajectory prediction (87% accuracy), and Multi-Armed Bandit optimization.
- Deployed via Streamlit with real-time inference at 45 FPS; published in IJCER 2024, presented at St. Francis College, Bangalore.

DriftAI: Drift Angle Prediction Dashboard

- Built a predictive ML model to forecast drift angles for motorsport vehicles with an interactive Streamlit dashboard. Tech: Python, Scikit-Learn, Streamlit, Pandas.

LuxeStay Royal Hotel & Dining — Hotel Booking System

- Built a full-stack hotel booking application using React, Vite, and Supabase for backend integration (PRCBO-2026).

IoT & Smart Security System (Research Paper)

- Designed an IoT-based smart security model with sensor-driven anomaly detection; presented at the Advanced Computational Intelligence & Disruptive Technologies Conference, PSG College, Coimbatore.

AREAS OF INTEREST

• Physics-Informed Machine Learning	• Deep Learning & Neural Networks
• Predictive Analytics & Time Series Forecasting	• Computer Vision & Object Detection
• Natural Language Processing	• Big Data Engineering
• Motorsport Data Analytics	

CERTIFICATIONS

- Data Visualization: Empowering Business with Effective Insights — TATA
- Data Science 101 (DS0101EN) — IBM
- Diploma in MySQL and Statistics for Data Analysis
- Data Science & Analytics — HP (2024)
- IC38 Insurance Agent Licensing Exam — Passed (license issuance pending)